

Document Control

- System Name: Application
- Version: 1.0
- Date: Not present in source JSON
- Owner: Not present in source JSON

Executive Summary

- This BRD outlines the ingestion of customer and order CSV files into Delta tables, implements data cleansing, computes order-level spend, joins datasets to create an order summary with SCD Type 2 change tracking for customer data, and generates a table with daily customer spend aggregates.
- Implementation is performed using Delta Live Tables, maintaining the exact names, paths, schemas, and business logic specified.

Business Objectives

- Ingest customer and order CSV data into designated Delta tables.
- Cleanse the ingested data by removing "Null" values and duplicates.
- Enrich order records with a computed TotalAmount (PricePerUnit multiplied by Qty).
- Join customer and order datasets on CustId to produce ordersummary.
- Implement SCD Type 2 logic for tracking customer attribute changes.
- Aggregate daily customer spend for analytics.
- Orchestrate all steps leveraging Delta Live Tables.

In-Scope

- Loading designated customer and order source files into Delta tables.
- Schema definitions and table creation for customer, order, ordersummary, and customeraggregatespend.
- Null value and duplicate removal.
- Addition of the computed TotalAmount column.
- Customer/order join processes with SCD Type 2 handling.
- Aggregation of spend results.
- Use of Delta Live Tables for all pipeline steps.

Out-of-Scope

- Data sources or systems not specified in the requirements.
- Access controls, SLAs, monitoring, and alerting.
- Performance tuning or scaling.
- User interfaces, dashboards, and reports.

- Scheduling, orchestration, or DevOps concerns beyond DLT implementation notes.
- Data quality metrics apart from listed cleansing steps.

Stakeholders

- Information not present in source JSON.

Business Requirements

BR-ING-001 — Customer and Order Ingestion

- Description: Load customer and order CSV files from specified paths into target Delta tables.
- Business Value: Centralizes raw operational data for reliable processing and analysis.
- Priority: High

BR-DCL-002 — Data Cleansing

- Description: Remove all "Null" (case-insensitive) and duplicate records from customer and order datasets.
- Business Value: Improves data quality and analytic accuracy downstream.
- Priority: High

BR-ENR-003 — Computed TotalAmount

- Description: Add a TotalAmount column to order table as PricePerUnit multiplied by Qty.
- Business Value: Enables granular spend analytics and aggregation.
- Priority: Medium

BR-JOI-004 — Customer–Order Join

- Description: Join cleansed customer and order Delta tables on CustId to form the ordersummary table.
- Business Value: Correlates order activity with customer attributes for reporting and change tracking.
- Priority: High

BR-SCD-005 — SCD Type 2 Handling

- Description: Track and maintain full history of customer attribute changes in ordersummary via SCD Type 2 (set previous records to Inactive, new records to Active, update StartDate/EndDate).
- Business Value: Preserves customer history and ensures data temporal accuracy.
- Priority: High

BR-AGR-006 — Customer Spend Aggregation

- Description: Aggregate ordersummary to create customeraggregatespend table, grouping by Name and Date, with sum of TotalAmount.
- Business Value: Provides enterprise spend analytics capability.
- Priority: Medium

BR-IMP-007 — Delta Live Tables Implementation

- Description: Implement requirements using Delta Live Tables, preserving steps, names, and schemas as provided.
- Business Value: Ensures manageability, scalability, and repeatability of the data pipeline using DLT capabilities.
- Priority: High

Assumptions & Constraints

- Only explicit requirements from the specification are used.
- Delta Live Tables is required as the implementation approach.
- Field names, case, and paths are preserved as provided.
- Additional SCD Type 2 attributes such as StartDate, EndDate, and Active flag are referenced but not fully defined.
- No external dependencies or attributes are considered.

Success Metrics

- 100% of records from specified sources are loaded without "Null" or duplicate contamination.
- Computed TotalAmount is present in order and customeraggregatespend tables.
- ordersummary contains complete customer and order attributes with SCD Type 2 history.
- Spend aggregation matches raw data roll-up post cleansing and SCD application.
- Delta Live Tables execution completes without pipeline errors.

Risks & Mitigations

- Ambiguity in SCD Type 2 attribute field types or values can affect correct history tracking; recommend confirming field definitions.
- Data quality controls are limited to Null and duplicates; edge cases (type mismatches, outliers) are not addressed.
- Error handling, monitoring, and alerts are not defined; risk of silent failures or data quality slip.
- Mitigation: Clarify SCD field requirements, validate pipelines with representative data, and extend scope to include QA if needed.

Approval

- Approver Name: Not present in source JSON
- Date: Not present in source JSON

- Signature: Not present in source JSON

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into Delta table customer
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into Delta table order

Appendix B: Target Definitions

	Layer	Table	Columns
	Bronze/Silver equivalent	customer	CustId, Name, EmailId, Region
	Bronze/Silver equivalent	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId
	Silver/Gold equivalent	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date
end	Gold	customeraggregatespend	Name, TotalAmount, Date

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Data Cleansing	customer/order	customer/order	Remove "Null", remove duplicates
2	Computed Column	order	order	TotalAmount = PricePerUnit × Qty
3	Join	customer, order	ordersummary	Join on CustId, SCD Type 2 logic
4	Aggregation	ordersummary	customeraggregatespend	Sum(TotalAmount) by Name and Date

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-1	Computed Column	TotalAmount = PricePerUnit × Qty	Requirements
BR-2	Cleansing	Remove "Null"/Null, remove duplicates	Requirements
BR-3	Join	Join on CustId	Requirements
BR-4	SCD Type 2	Maintain historical records, Inactive/Active flags, StartDate/EndDate updates	Requirements
BR-5	Aggregation	Group by Name, Date, sum TotalAmount	Requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Ingest customer/order files to Delta tables	Source files present
2	Cleanse data (Null/duplicate removal)	Step 1
3	Compute TotalAmount in order	Step 2

4	Create ordersummary via join and SCD Type 2	Steps 2-3
5	Aggregate spend to customeraggregatespend	Step 4

Process Flow Diagram

